

Tender Submission

Submitted by: Esri China (Hong Kong) Limited

Submitted to: Department of Land Surveying and Geo-Informatics (LSGI), The Hong Kong Polytechnic University (PolyU)

Title: Design of a Robust Algorithm to Quantitatively Evaluate Greenspaces Accessibility and Livability in Hong Kong: A People-Centric Approach

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Introduction

In Hong Kong, greenspaces can be important in enhancing environmental and urban residential living quality, by improving the air quality, having open spaces with physical activities, positive impacts on mental health, and promoting biodiversity. However, Hong Kong would require a systematic approach in collecting the greenspaces data and evaluating greenspace accessibility at neighborhood level, especially in densely populated areas. Therefore, Esri China (Hong Kong) is committed to the project of collecting greenspaces data and developing an algorithm to evaluate livability with statistics, by using advanced geographic information systems (GIS) software and skills in geospatial technologies. This submission describes our comprehensive approach of developing a robust algorithm and scoring model in a people-centric approach.

Objective

This project aims to develop a greenspace scoring index model using high-resolution geospatial data at neighborhood level. This model will reflect the quality, accessibility, and distribution of greenspaces across Hong Kong. The algorithm aims to provide recreational GIS information to the public for accessibility, enhanced greenspace allocation and development in urban planning, ultimately enhancing livability.

Company Background

Esri is a global company and specializes in GIS software development and geospatial analytics and over 50 years of experience. Our ArcGIS platform is used worldwide for mapping, analysis, and visualization of geospatial database. Esri China (Hong Kong) has adapted the software to the local data, with experience in geospatial data collection, processing, and analysis. Our team uses a wide range of surveying equipment for flexibility and efficiency, such as UAV photogrammetry, satellite imagery and LiDAR. We have applied mobile mapping systems equipped with LiDAR for data collection for ArcGIS base map with success. Therefore, we believe our team in Hong Kong is well-equipped to carry out the solutions for evaluating green space accessibility in Hong Kong.

Tender Document Details:

- a. Project team
- b. Solution proposal
- c. Deliverables
- d. Timeline management
- e. Budget proposal

Project team

1. Key Personnel

To ensure successful execution of this project, Esri China (Hong Kong) has the following guidelines acquiring experts for the project team, with the relevant skills and responsibilities in geospatial technologies and geodatabase analysis:

- **GIS Analysts:** Analyzes geospatial data, ability to present data in visualized diagrams and geoprocessing software such as ArcGIS Pro. They shall also have knowledge processing satellite images and point cloud data from UAV photogrammetry and LiDAR scanners, experience in 3D modeling. They shall hold degrees from Land Surveying and Geoinformatics (LSGI). In a typical GIS project, they serve as the GIS project manager, database manager, and digital map maker. They group and integrate GIS data into common formats for the handover to the data scientists.
- **Surveyors:** Operates surveying equipment such as backpack and static LiDAR scanners, scanning mobile vehicles, and UAV drones. They focus on raw data acquisition, capturing 3D point cloud data of the pedestrians and roadside features. They shall hold LSGI degrees, with a driving license and/or Category A2 SUA license is an advantage.
- **Data scientists:** Responsible for the programming of the algorithm to evaluate greenspace accessibility. They will focus on algorithm development, scoring sheet design, and using machine learning to account for the factors such as the vegetation index and implementing algorithms such as the nearest neighbor problem to create Green Space Index (GSI). They shall hold Computer Science (CS) degrees. In a typical GIS project, they serve as the system operator and the programmer.
- **Project managers:** Oversee the project development and timeline management. They shall have no less than 5 years of experience supervising geospatial projects and holds LSGI or CS degrees. They will lead this project and capable to serve as interims for GIS analysts, surveyors, or data scientists. In this project they shall also be responsible for communication with Prof. Zhu Xiaolin from LSGI and report the team's progress.

2. Team members

The proposed team consists of 15 members, the team member composition can be summarized with the following table:

Role	Count
Project managers	2
GIS Analysts	4
Data scientists	4
Surveyors	2
Chainmen	2
Drivers	1

Solution proposal

a. Raw data acquisition

We will use various surveying equipment to collect the raw data in point cloud and photogrammetry. The vehicles equipped with LiDAR scanners can collect the data for the road side greenspace features and vegetation areas to a larger extent. The backpack LiDAR scanners can collect smaller extent, people-centric data, such as pedestrian areas, rooftop gardens, and parks. UAV photogrammetry can provide efficient data collection for forming the 3D model, including the height of the data, providing localized vegetation information at a large scale. We also use satellite images from USGS Landsat which can provide land use information, vegetation indices such as the NDVI for small scale, very large extent areas.

The approach of using remote sensing and photogrammetry such as LiDAR scanning and UAV drones reduces the disruption to the neighborhood, since noise is especially sensitive in greenspace areas. Our data collection methods also comply with the Hong Kong's land surveying and safety ordinances, for example, the pilot shall have the Category A2 license for SUA, and drivers shall have driver's license. As for the photos and point clouds collected, we will comply with the Hong Kong's data privacy regulations.

b. Scoring model development

We will use the raw data to develop a greenspace scoring index (GSI) that measures the accessibility and livability of the people using people-centric approach, using multiple factors.

For accessibility, the raw data will be processed and generate a path for people to get to the greenspace areas using weighted heuristics such as the sidewalks, roads, highways. The 3D model also assesses the terrain of the areas, whether they are flat or rugged. The areas can be grouped using K-nearest neighbor algorithm to find the nearest consecutive greenspace areas and form clusters of different sizes. The shortest paths for the people can be found using the A-star algorithm. Overall, sizes of the greenspaces, density, distances can define the accessibility of the people to the greenspaces.

For livability, the NVDI index from the satellite images can indicate the vegetation health by processing the near infrared (NIR) channel and the red channel using the NVDI formula. The point cloud data can be applied with a computer vision algorithm to automatically identify the recreational facilities (benches, playgrounds). Overall, the NVDI index and the recreational facilities can define the livability of the greenspaces.

The above factors will be weighted in the final algorithm, according to the literature review on week 13, stakeholders from the using GIS software for optimization

c. Deliverables

- GIS platform: Viewing the 3D map formed by UAV photogrammetry and point cloud data. Functions as a proof of concept and result query of the GSI algorithm.
- GSI algorithm: Directly querying the GIS platform for GSI analysis and GIS platform can display the heatmaps.

d. Timeline management

The project is expected to take 6 months, from 15 Oct 2019 to 10 Apr 2020, aligning with the requirements in the original EOI. The following is the proposed timeline:

Week 1-8: UAV drone surveying with photogrammetry and mobile LiDAR scanning, combined with backpack LiDAR scanning for pedestrian and parks; Data analysis and import into GIS software; Point cloud processing for the UAV photos and form 3D mesh model.

Week 8-13: Development and refinement of GIS platform; Development of GSI algorithm and initial prototype, using a people-centric weighted approach; Point cloud processing for LiDAR scanning data to form 3D mesh model.

Week 13 (3 Jan 2020): Submission of the Progress Report, including the completed tasks, 3D mesh models, development of GIS platform, and preliminary findings.

Week 14-20: Testing and debugging of the GSI algorithm, and the GIS platform on the user end. This phase validates the platform and ensures it functions normally.

Weeks 21-24: Deployment of GSI algorithm on the GIS platform; Testing and refinement of the GSI algorithm, validating the greenspace accessibility and livability for its accuracy.

Week 25 (1 Apr 2020): Final scoring model submission, delivering a fully functional algorithm.

Week 25 (3 Apr 2020): Final presentation with PowerPoint slides, summarizing the raw data acquisition and scoring model of GSI, and accessing the analyzed data through the GIS platform.

Week 26 (10 Apr 2020): Final report including introduction, methodology, results, and conclusion. Detailed descriptions for deliverables such as the GIS platform equipped with GSI algorithm.

Budget Proposal

Personnel Costs:

Role	Count	Daily Wage	Days per week	Weeks	Total Cost (HK\$)
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Project managers	2	1200	5	26	312 000
GIS Analysts	4	900	5	8	144 000
Data scientists	4	900	5	18	324 000
Surveyors	2	900	5	8	72 000
Chainmen	2	700	5	8	56 000
Drivers	1	700	5	8	28 000

Total Personnel Costs: HK\$ 1 232 500

Total Budget Proposal

Item	Estimated Cost (HK\$)
Personnel Cost	936 000
Computer hardware	30 000
GIS Software by Esri	0
Bentley software license	35 000
Backpack and mobile LiDAR scanners	500 000
UAV drone	10 000
Traveling allowance	20 000
Miscellaneous	50 000

Total estimated cost: HK\$ 1 581 000

Conclusion

Esri China (Hong Kong) has the experience and expertise in GIS software development and geospatial data processing and mapping, can ensure us to successfully deliver a robust, people-centric algorithm for greenspace index evaluation for accessibility and livability. Esri believes that this solution provides professional surveyed data and carefully designed scoring model, ultimately providing insights for public information and urban land use planning. We look forward to collaborating with LSGI to deliver this solution, benefiting the general public and the decision makers through GSI.

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